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Optimal investment and operational planning of a storage power plant

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ABSTRACT

Liberalization of electricity markets and dissemination of renewable energy sources leads to the variability of electricity prices, intermittency of generation and thus the necessity to develop new electrical energy storage systems for integration of renewables. Due to the high volatilities, stochastic optimization methods need to be applied for operational and investment planning of power plants. This paper presents a stochastic approach to solving a combined optimization problem for short-term scheduling and long-term investment planning of storage power plants for 30-year long planning horizon. The proposed methodology is applied to cost-benefit analysis of stand-alone storage plants operating under market conditions through price arbitrage: pumped hydropower plant and hydrogen storage plant. The results show that the pumped hydropower plant is profitable, whereas the hydrogen storage plant project is not economically viable without additional support. Possible economic incentives resulting in a profitable hydrogen storage plant are examined.

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Introduction

Background of the problem

Liberalization of the power sector requires a significantly revised approach to both long- and short-term operational planning of a generating company (GENCO[1]). The GENCO's profit is subject to significant fluctuations of energy market prices, fuel cost, ambient temperature, resource availability such as water inflow to hydropower plant (HPP) reservoirs, wind speed, solar radiation, etc. The uncertain and variable parameters need to be taken into account not only for optimization of the daily bidding strategy and unit commitment decisions, but also for long-term investment planning of building new power plants.

To participate in the day-ahead electricity market (such as the Elspot of the Nord Pool power exchange), the GENCO submits its bids based on forecasts thus incorporating uncertainty into the daily operational planning. Once the power production bid is submitted to the market, it can be changed

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[1] Abbreviations: ANN: artificial neural network; BOP: balance of plant; GENCO: generating company; HPP: hydropower plant; IRR: internal rate of return; NPV: net present value; O&M: operation and maintenance; PCS: power conversion section; PSH: pumped-storage hydropower; RES: renewable energy sources; TTC: total capital costs.

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only with surplus payment. Uncertainty has become even more important for long-term investment planning and cost-benefit analysis of new power plant projects. Besides the variability based on market conditions and environmental considerations, uncertainties due to changes in legislation also need to be taken into account. Consequently, the problem of profit maximization of the GENCO shall be formulated so as to take into account the uncertain and random parameters. As a result, the problem becomes much more complicated than a deterministic one and requires new algorithms and software tools to solve it.

Fluctuations in the power generation arise as a result of dissemination of intermittent renewable energy sources (RES), the role of which is continuously growing due to the European Union renewable energy targets and incentives in other parts of the world. Consequently, large-scale energy storage becomes as important as ever before to make maximum use of the variable RES, introduce operational flexibility and assist in balancing the generation and load of the power system.

This research deals with stand-alone storage power plants operating in a market environment with the goal of profit maximization through price arbitrage. The storage plants are considered to be price-takers and electricity prices are exogenous. Investment planning and short-term operation optimization of storage power plants under day-ahead market conditions is researched in this paper. It can be considered as the pre-feasibility study of storage power plant projects.

Two options of energy storage are assessed: pumped-storage hydropower and hydrogen storage. The power plants under study employ power-to-power conversion. It is important to note that these two technologies usually cannot be regarded as competitors in respect to each other in a specific site, since each facility has its own requirements towards the location that only seldom are met simultaneously. The characteristics of both studied storage technologies are presented in the next subsection.

Energy storage

To allow the transition to a RES-based power system, large-scale energy storage is required for short-term and seasonal balancing [1], [2]. There is an ongoing extensive research on large-scale energy storage deployment in the existing power systems, since energy storage is considered as one of the solutions to avoid uneconomical power production and high peak prices. Energy storage can also diminish grid congestions and facilitate distributed generation [3].

Various energy storage options are being considered and used such as mechanical energy storage (hydropower plant reservoirs, pumped-storage hydropower, compressed air, flywheel), electrochemical battery energy storage (Lithium-ion batteries etc.), electric and magnetic energy storage (capacitors and supercapacitors, superconducting materials), power to gas storage technologies (with conversion to hydrogen and methane), thermal energy storage etc. [3]. In this study, two types of storage power stations are considered: pumped-storage hydropower plant and hydrogen storage power plant.

Pumped-storage hydropower

Pumped-storage hydropower (PSH) is the mostly used storage option in the power sector worldwide as it takes 99% of the electricity storage capacity and is the only commercially viable large-scale storage without additional fuel needs [3]. PSH is considered as the most suitable and efficient large-scale storage method [2] with new projects of PSH plants proposed lately in many countries as a result of development of RES and liberalized electricity markets [3]. However, the main restriction for this type of storage is the lack of suitable large land areas [2]. Besides, a recent research has established that PSH is more expensive for long-term storage than compressed air and hydrogen storage [4].

PSH plants can be pure pumping or mixed type when combined with a conventional hydropower plant with a reservoir [5]. In this research, a pure PSH plant is considered operating under market conditions and profiting through price arbitrage from hydro generation during peak-price periods using the water pumped during off-peak hours.

Hydrogen storage

In contrast to the PSH which is a mature technology and the oldest of all the available energy storage types [5], hydrogen is promoted as the future energy carrier in many studies. It is concluded that hydrogen has the highest potential to serve as alternative to fossil fuels in terms of availability and low emissions of greenhouse gases [6]. Moreover, hydrogen has a capability of storing energy in various forms, such as gaseous form, liquid form or in the form of metal hydrides, and it can be suitable both for small- and large-scale storage [6]. In general, hydrogen can be produced from a wide range of resources using various energy conversion processes [2]. Due to the various hydrogen storage forms and the broad application possibilities, hydrogen can serve as a link to combine the electricity, transport and heating sector in one energy market [1].

Electricity is converted into hydrogen via water electrolysis, a process often referred to as “power to gas”. To implement the opposite process, called “gas to power”, fuel cells can be used to generate power and heat through electrochemical reaction of hydrogen and oxygen. Depending on their type, fuel cells can be employed for micro combined heat and power generation (micro-CHP) locally or for larger-scale electric supply [7], [8]. Another important application area of fuel cells is for backup power where fuel cells have much longer service life compared to battery-based backup [8]. Moreover, fuel cells are widely considered for usage in transportation in competition with battery electric vehicles. Thus, fuel cell technologies can serve as one of the components for an integrated energy system consisting of electric grid, thermal grid and fuel grid. It is planned to base the future fuel grid on hydrogen as the energy carrier [8], produced from water via electrolysis and used for vehicles powered by fuel cells or hydrogen engine to replace gasoline, increase fuel use efficiency and reduce greenhouse gas emissions [9]. Thereby, the transportation system can also act as a utility-scale energy storage to enable dispatch of energy between the various parts of the integrated energy system, to improve grid reliability and promote RES integration [8].

For large-scale and long-term hydrogen storage, underground storage (e.g. in depleted gas and oil fields, aquifers, salt caverns) is assessed as a promising storage type [1], [9]. It is considered as one of the alternatives to large-scale pumped-storage hydropower [2], [4]. Hydrogen can be produced from water and electricity and converted back to electricity in combined cycle power plants. Though the electricity-to-electricity conversion efficiency of this process is less than 40%, underground hydrogen gas storage is currently the only method with a technical potential for storing energy at capacity of 100 GWh with time scale of weeks [2]. Sites suitable for this type of storage can be found in many parts of the world. For example, in Latvia there are at least eleven potential underground gas storage sites containing porous sandstone with up to 50 billion m^3 of total active storage capacity. One of the sites—Incukalns—serves as the third largest natural gas storage in Europe [10] and has been in operation for several decades.

In our case study, hydrogen storage plant with power-to-power conversion is considered using a regenerative system where hydrogen is produced via electrolysis during off-peak hours and fuel cells are employed for electricity generation at the peak price.

The main contributions of the paper are the proposed mathematical statement of the problem which allows to take into account uncertainties, ANN-based forecasting of the volatile parameters and the optimization algorithm which includes long-, medium- and short-term planning and allows to perform optimal operational and investment planning.

The reminder of the paper is organized as follows. 2nd section presents the forecasting approach used for prediction of the day-ahead electricity market price. Next, the model for power plant feasibility calculation and fine-tuning of the model is presented. The 4th section provides results of numerical simulations carried out based on the proposed forecasting and optimization approach for optimal investment and operational planning of storage power plants. In the final section conclusions are drawn.

Forecasting of the electricity market price

In order to take into account uncertainties due to market conditions, it is necessary to forecast the variable hourly day-ahead electricity market price. We perform predictions using artificial neural networks (ANN). Initially, point forecasting is performed and then prediction errors are considered by imposing those onto the point forecasts. Thus numerous realizations of price forecasts for each day under interest are obtained as further described in this section. These realizations are then used to account for stochasticity in the optimization procedure.

Main assumptions

Initially, let us assume that:

- All the predictions can be performed by forecasting the mean value of variables (point forecasting)
- Historical forecast data and prediction errors (calculated after observations of the actual data) have been logged and are available
- Statistical properties of prediction errors are constant in time allowing us to generate the realizations of stochastic processes. The duration of those processes is limited only by the available historical data and the rate of their aging.

Based on the assumptions above, we use a naïve model [11] for estimating prediction errors (Fig. 1). As a result, when the day-ahead prediction of the mean values is made, it is adjusted by adding the historical forecast errors, thus allowing to simulate realizations of the future processes.

ANN-based point forecasting

Due to the ability of ANNs to map nonlinear complex ties within a multidimensional information domain, a large number of studies employ them in forecasting. Previous studies [12,13] present the key guidelines on how to make and evaluate point forecasts particularly by using a multilayer feedforward ANN for time series prediction [12].

The main issues in the application of ANNs are related to maximizing the credibility of prediction. To properly address those issues, the parameters of the ANN and the related procedures should be carefully chosen. It includes the network architecture (type of ANN, number of hidden layers and neurons in the hidden layers); activation functions; selection of the training data and the size of training data sets; preprocessing of data; the number of training cycles; learning and validation algorithms [12–14].

In our study, we have selected a three-layer feedforward ANN and we employ the Levenberg–Marquardt training algorithm with backpropagation. The choice of the architecture and training algorithm of the ANN was largely defined by the tools available in the MATLAB Neural Network Toolbox and results of other studies [12,13].

Training of the ANN is performed in a supervised manner using the following hourly historical data of 2013–2015 for input (Fig. 2):

- Day-ahead electricity market price of the Nord Pool Latvian bidding area
- Day of week and hour
- Ambient temperature.

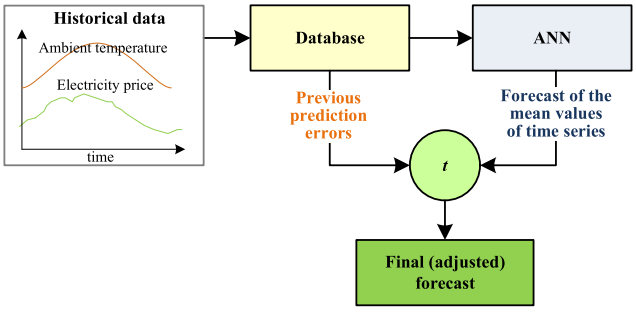


Fig. 1 – Representation of the naïve model for estimating prediction errors and adjustment of the forecast.

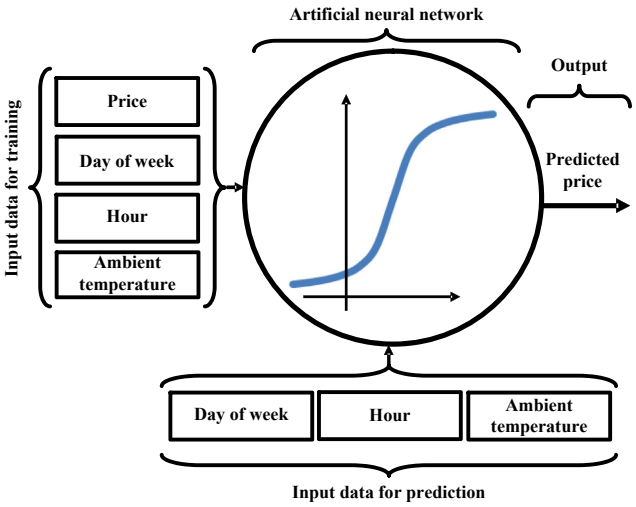


Fig. 2 – Input data for training the ANN and forecasting electricity price.

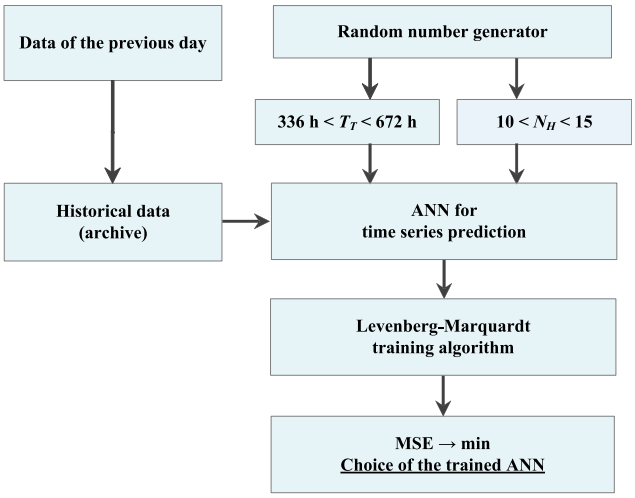


Fig. 3 – A simplified diagram of the algorithm for choosing the ANN parameters.

For prediction of the day-ahead price, the following data is input to the trained ANN:

- Day of week and hour
- Ambient temperature.

The output of ANN provides hourly forecast of the day-ahead electricity market price which is then used as basis for power plant feasibility calculation. Additionally, for the long-term cost-benefit analysis of storage plants, an annual market price increase is assumed as presented later in our case study.

During the training of ANN, we observed a high sensitivity of prediction accuracy to the number of neurons in the hidden layer and the length of historical time series used for training. These trends can be explained by seasonal changes of the probabilistic properties of processes under study. Therefore we made a decision to select some of the parameters of the ANN automatically immediately after the receipt of new data, before each new prediction is conducted.

We have designed an algorithm to specify the most suitable parameters of the ANN (Fig. 3), namely, the length of the training data set, T_T, and the number of neurons in the hidden layer, N_H. A partial enumeration is performed by randomly selecting the above parameters within the specified limits (T_T from 336 to 672 h and N_H from 10 to 15 neurons) to evaluate the forecast performance for the previous day with the selected parameters. The minimum prediction error of the forecasts for the previous period is used as the criterion for selecting the most suitable ANN parameters for the next forecast horizon. Competing forecasts made by ANNs with differing parameters are compared to the actual observed data and assessed by means of an error measure, namely the mean square error: $MSE = 1/T \sum_{t=1}^{T}(x_t - y_t)^2$, where x_t are the point forecasts made for the previous forecast period $T = 24$ h, and y_t are the corresponding actual observations. After such a partial enumeration, the ANN parameters that have provided the least MSE are chosen for the next forecast horizon. This procedure is repeated for each day of the forecast.

Mathematical model for storage plant feasibility calculation

During this study we have developed a model to calculate the economic feasibility of a closed-loop energy storage plant that buys and sells electrical energy in a spot market. This model can be applied to any full cycle storage project regardless of the energy storage medium for as long as the technology can be characterized by the following parameters: energy accumulation efficiency, energy discharge efficiency, total installed capacity, discharge time and capital cost for storage plant construction. In our model the storage power plant is considered as a stand-alone plant.

General algorithm for optimization

The main purpose of the model is to estimate the annual expenditure and revenue of storage plant projects during their lifetime. These parameters can then be used to calculate the net present value (NPV), payback period and internal rate of return (IRR) for feasibility assessment.

For an electrical energy storage plant that operates by price arbitrage, accurate price predictions are crucial. However, since perfect forecast is impossible, the stochastic nature of spot prices has to be taken into account. In this study, we implement it by considering several realizations of hourly electricity price forecasts for short-term scheduling. Then the short-term optimization results are generalized to long-term employing time-averaging technique and thus combining long-, medium- and short-term planning as presented in the structure of algorithm in Fig. 4.

1. As input data we use the technical and financial parameters of the project as well as price statistics and predictions. Then in the first block of the algorithm a certain number of days (N) is randomly selected for each year of the planning horizon.
2. For each of the randomly selected N days, deterministic medium-term optimization is performed using linear

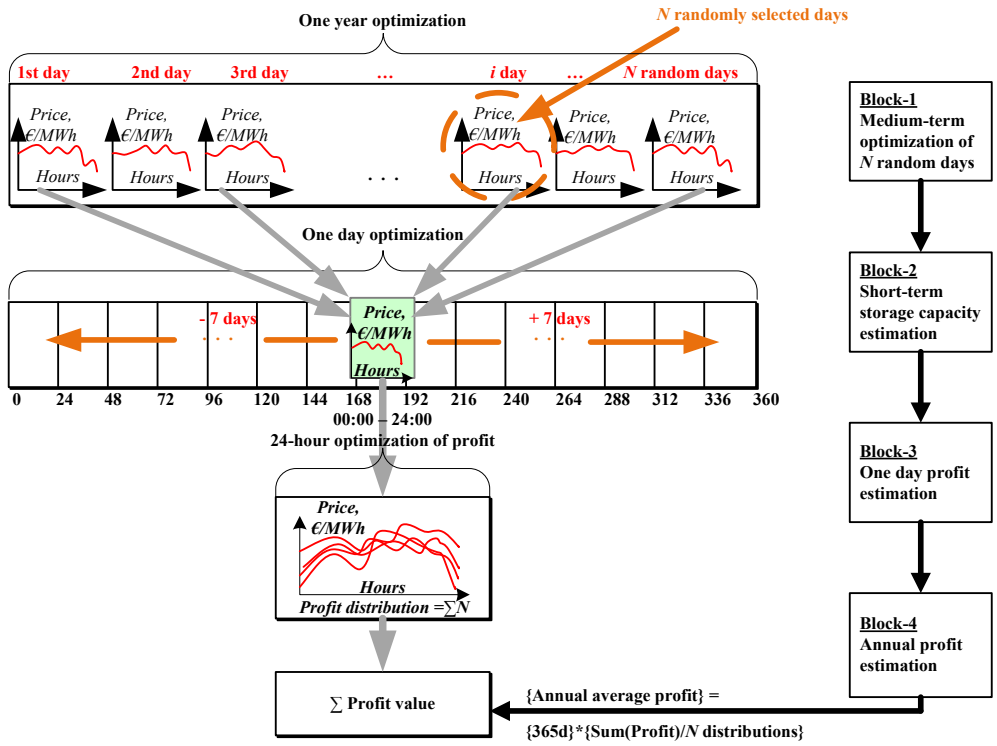


Fig. 4 – General algorithm for combined optimization of long-, medium- and short-term planning.

programming which involves optimization of one or two weeks before and after the selected day in hourly resolution. Linearization of the problem at this stage produces an insignificant error of approximation [15], since it provides only the optimal filling level of the storage system (e.g., reservoir level) at the beginning and end of the selected day.

3. Afterward short-term (24-h) optimization is done for each of the selected days using nonlinear programming to obtain the expected daily income and expense.
4. The results are then generalized for the whole year to calculate the NPV based on the assumption that the expected annual net cash flow, *NCF*, can be computed as the time-average value [16]: $E[NCF] \approx \overline{NCF} = \lim_{T\to\infty} \frac{1}{T} \int_0^T NCF(t)dt$. For this assumption to be valid, the length of the time-averaging interval needs to tend to infinity. Practically it can be chosen based on the requirements for accuracy, computational expense and complexity of the problem represented by the integrand. This is further discussed in subsection Number of Monte Carlo trials in a year where the necessary number of days for sampling is selected.

Mathematical description of the model

The optimization variable is the change of stored energy at every hour, $\Delta L_{i,t}$. The objective function (1) (sum of the products of the hourly generated or consumed power and its price) is piece-wise and non-smooth, which means that conventional nonlinear solvers that use gradient methods have no guarantee of finding a solution. *M* realizations of price forecasts are considered to account for uncertainties in short-term scheduling (for medium-term optimization $M = 1$):

$$f_i(\Delta L) = \frac{1}{M}\sum_{m=1}^{M}\sum_{t=1}^{T} P_{i,t}\cdot c_{i,m,t} \to \max \quad \forall i \in N \tag{1}$$

for

$$\begin{cases} P_{i,t} = -\Delta L_{i,t}\cdot k/\eta_{rt} & \text{if} \quad \Delta L_{i,t} > 0 \\ P_{i,t} = -\Delta L_{i,t}\cdot k & \text{if} \quad \Delta L_{i,t} \le 0 \end{cases} \tag{2}$$

subject to

$$\sum_{t=1}^{T} \Delta L_{i,t} = L_T - L_0 \tag{3}$$

$$-\sum_{t=1}^{S} \Delta L_{i,t} \le L_0 - \underline{L} \tag{4}$$

$$\sum_{t=1}^{S} \Delta L_{i,t} \le \overline{L} - L_0 \tag{5}$$

where:

$i \in N$ indices for the *N* randomly sampled days;
$t \in T$ hours;
$m \in M$ price realizations;
$P_{i,t}$ consumed or generated power at hour *t*, day *i*;
$c_{i,m,t}$ hourly predicted price;
$\Delta L_{i,t}$ change in the amount of stored energy at hour *t*, day *i*;
k energy to power conversion coefficient depending on the storage technology;
η_{rt} round trip efficiency;

L_0, L_T the initial and final amount of stored energy;
$\underline{L}, \overline{L}$ the lower and upper bound on stored energy;
$S \in T$ variable to enforce storage capacity bounds at each hour.

The model is implemented in MATLAB R2013a scripting environment utilizing Optimization Toolbox 6.3 and Global Optimization Toolbox 3.2.3 [17]. Since the objective function is not twice continuously differentiable, we employ pattern search algorithm [18] to find the optimal solution. To increase the speed of optimization, the initial point for the pattern search algorithm is supplied by constrained nonlinear optimization using interior–point method.

Once the values of annual revenue and expenditure on purchased electricity are obtained, they are used in calculating the net present value (NPV) of the studied project.

NPV is an often used indicator for deciding whether a project or investment will result in a net profit or a net loss. A negative NPV results in loss, whereas a positive NPV results in profit. NPV of a project is defined as the difference between the present value of its benefits and the present value of its costs over a period of time. It can be expressed as [19]:

$$NPV = \sum_{y=0}^{Y} \frac{NCF_y}{(1+r)^y} = \sum_{y=0}^{Y} \frac{R_y - (I_y + OM_y)}{(1+r)^y} \tag{6}$$

where:

y time of the cash flow (year);
Y number of periods in project's lifetime (in our case—30 years);
r discount rate;
NCF_y net cash flow in year y;
I_y investment cost in year y;
OM_y revenue in year y;
OM_y operation and maintenance (O&M) cost in year y.

Annual revenue is estimated by employing time-average approach and taking into account several price realizations:

$$R_y = \frac{365}{M \cdot N} \sum_{i=1}^{N} \sum_{m=1}^{M} \sum_{t=1}^{T} P_{i,t} \cdot c_{i,m,t} \quad \text{for} \quad P_{i,t} > 0 \tag{7}$$

Annual O&M costs consist of fixed O&M and variable expenditure on purchased electricity for energy accumulation. Additional variable O&M costs are not taken into account at this study as they constitute a relatively small amount.

$$OM_y = OM_{\text{fixed}y} + \frac{365}{M \cdot N} \sum_{i=1}^{N} \sum_{m=1}^{M} \sum_{t=1}^{T} P_{i,t} \cdot c_{i,m,t} \quad \text{for} \quad P_{i,t} \le 0 \tag{8}$$

where:

$OM_{\text{fixed } y}$ annual fixed O&M costs

Finally, the third component necessary to calculate the net cash flow and NPV (6) is the investment cost. In order to account for contingencies in the construction phase we add a certain percentage to the capital costs to estimate the total investments.

$$I_y = (1 + d_{\text{cont}}) \cdot TCC \tag{9}$$

where:

d_{cont} project contingency cost;
TCC total capital costs (cost components considered in this paper are further detailed in the case study).

Internal rate of return (IRR) is another index used to evaluate investment opportunities. IRR is equal to the discount rate which makes the NPV of a project equal to zero by the end of its lifetime. In other words, it is the maximum interest rate affordable for the project to repay its investments and break-even. The IRR is obtained by solving the equation:

$$\sum_{y=0}^{Y} \frac{R_y - (I_y + OM_y)}{(1+IRR)^y} = 0 \tag{10}$$

Tuning of the model

Before the model could be used for feasibility studies, some of its main parameters were adjusted to ensure reliable results. First of all, it was necessary to decide on the length of the medium-term planning horizon and the state of the storage system at the beginning and at the end of the planning period. Secondly, the number of Monte Carlo simulations in a year had to be chosen. Larger number of trials leads to higher precision of calculations, while it also increases the computational burden. Thirdly, the number of electricity price forecast realizations for a single day needed to be selected. Finally, the effect of performing short-term optimization using several realizations of forecasted processes taking into account possible prediction errors (as described previously) as opposed to one point prediction had to be studied.

This subsection analyzes the above-mentioned considerations for fine-tuning of the model.

Length of the medium-term planning horizon

Two potential medium-term planning horizon lengths are considered: 360 h (15 days) and 696 h (29 days), i.e. 1 or 2 weeks before and after the randomly selected day for short-term optimization. Decreasing the length of the medium-term planning period allows to diminish the computational effort, but it is permissible only if it does not have a significant effect on the storage state and generated revenue on the day considered in the short-term planning. For this purpose a test case was developed. We performed short-term optimization on several separate days using actual prices from the Nord Pool Spot and a hypothetical storage plant with 25 MW of installed power, round trip efficiency 0.72 and discharge duration 6 h. The input data (initial and final amount of stored energy) were taken from the corresponding medium-term planning stage.

Results discussed below present the case which produced the most notable data discrepancy between the two considered time periods.

Short-term optimization for scheduling of the power plant on January 20, 2015 was performed considering two different medium-term horizons: January 13–27, 2015 (15 days) and January 6 – February 3, 2015 (29 days).

As shown in the scheduling results for the 29-day long planning horizon (Fig. 5), the optimal initial and final storage level for short-term optimization should be 40.05% and 26.91%. This results in 4059.58 euro profit for January 20, 2015 (Fig. 6).

On the other hand, the parameters obtained from the 15-day horizon were 40.06% of initial storage, 27.29% of final storage and 4025.56 € profit from short-term optimization of the same day.

The above data show that by decreasing the medium-term scheduling horizon from 29 days to 15 days, there is an insignificant change of the optimum filling of storage at the beginning and end of the day for short-term optimization. As a result, the profit on the particular day decreased by only 0.84%. Thus, we can conclude that the medium-term planning horizon length of 360 h (15 days) is sufficient and will be used in all the further calculations.

Another aspect of the medium-term planning stage to be considered is the effect of the initial and final state of the 15-day period on the short-term period. We performed optimization for three different cases for the previously selected time period. In the first (reference) case, the initial and final storage fulfillment was equal (50%); in the next case storage fulfillment decreased from 100% to 0%; and, finally, it was set to increase from 0% to 100%. The second case resulted in increase of the short-term profit by 0.16% and the third case—by 0.91%. Thus it can be concluded that the initial and final state of storage in the medium-term horizon does not have a significant effect on the short-term optimization. Consequently, all the further calculations will be performed assuming storage fulfillment level of 50% both at the beginning and at the end of any 15-day period.

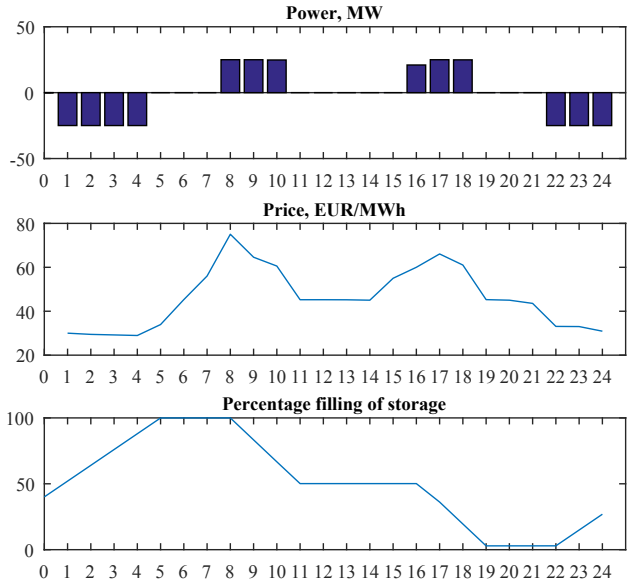


Fig. 6 – Short-term optimization results for January 20, 2015.

Number of Monte Carlo trials in a year

Short-term scheduling is performed for a number of yearly selected days and serves as the basis for long-term planning. To find an appropriate number of Monte Carlo trials that does not considerably decrease the accuracy and, at the same time, does not increase the computational burden too much, we performed several simulations using different numbers of trials in a year. The reference case is 365 trials, i.e. sampling each day of the year. The rest of the simulations (2, 5, 10, 20, 40, 80 and 160 trials) are compared to the reference case to determine the sampling error. The results are summarized in Fig. 7.

All the calculations were performed on a PC with two Intel Xeon E5-2630 v3 processors @ 2.40 GHz and 16 GB of RAM. MATLAB Parallel Computing Toolbox was used which allowed performing calculations on up to 12 workers at once. We established that 80 trials is a sufficient number as

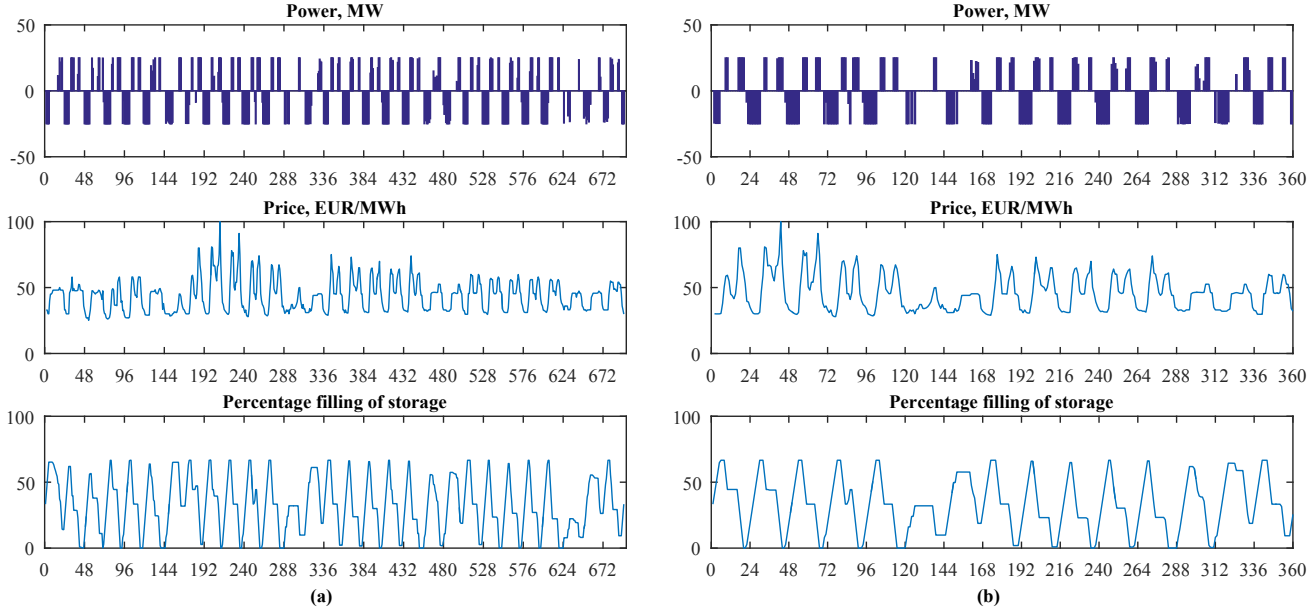


Fig. 5 – Medium-term optimization results with a planning horizon of (a) 696 h and (b) 360 h.

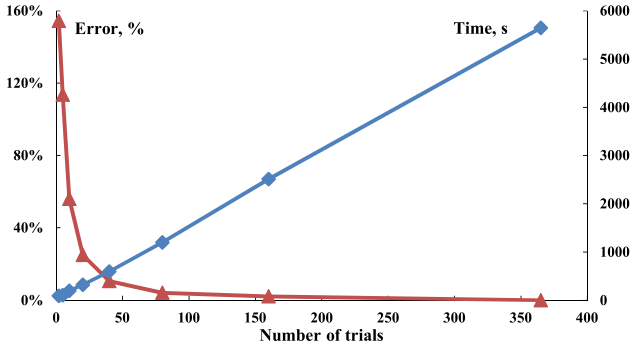


Fig. 7 – The time of computation and calculation error depending on the number of Monte Carlo trials.

it both did not introduce too high estimation error (4.1% for the test case) and did not require excessive computation time (19 min 58 s).

Stochastic versus one point prediction for short-term optimization

The final step in fine-tuning of the model was determining whether it is beneficial to employ stochastic approach when solving the 24-h planning subproblem by taking into account several price forecast realizations as opposed to only one point prediction which is used for medium-term optimization.

The process of obtaining point forecasts of electricity market price was described previously. By applying the noise generated by prediction errors of previous forecasts, we simulate additional forecast realizations [15].

It should be noted that for this test we used a total of 11 realizations. Predictions for January 21, 2015 (Fig. 8) were used in the short-term optimization algorithm. When all the 11 realizations were taken into account, the expected daily profit increased from 1727.2 to 1905.8 € providing an improvement of 10.34%. It can be concluded that it is beneficial to employ the stochastic approach to consider the uncertainty of price forecasts for short-term operation optimization of the storage plant. This will also reflect in the long-term planning when estimating the potential profit.

Case study

The model presented above was used to assess the economic potential of a hydro pumped-storage and hydrogen storage closed-loop plants. The main characteristics of power plants are summarized in Table 1.

As the basis for investment cost calculation we used the total capital cost (TCC) approach as presented in Ref. [3]. It includes the following cost components [3]:

- Power conversion section (PCS) costs: power interconnections (turbine/pump (PSH), fuel cells, converter, rectifier), cabling and piping
- Storage section costs: construction and excavation (cavern, reservoir)
- Balance of plant (BOP) costs: project engineering, grid connection and system integration, storage plant isolation and protective devices, construction management, land and access, buildings and foundation, HVAC system, monitoring and control systems, shipment and installation costs.

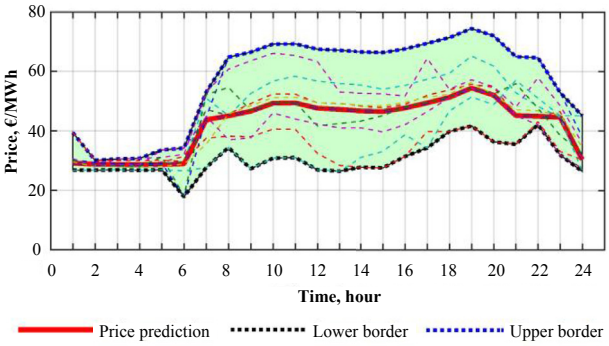


Fig. 8 – Electricity market price forecasts for January 21, 2015.

Table 1 – Technical and economic parameters of the storage plants under study.

Parameter	Technology	
	Pumped-storage hydropower	Hydrogen storage
Installed generation capacity	25 MW	25 MW
Generation efficiency	0.9 (turbines)	0.7 (fuel cells)
Installed storage capacity	150 MWh	150 MWh
Charging efficiency	0.8 (pumps)	0.7 (electrolysis)
Discharge time at full capacity	6 h	6 h
PCS and BOP costs [3]	528 €/kW	1383 €/kW
Storage section costs [3]	68 €/kWh	3.7 €/kWh
Total capital costs [3]	23 400 000 €	35 130 000 €
Fixed O&M costs [3]	4.6 €/kW	16 €/kW

Additionally, contingency costs are assumed to be 10% of the TCC for each of the projects.

Given 25 MW of power and 150 MWh of storage, the TCC are 23 400 000 € for the pumped-storage hydropower (PSH) plant and 35 130 000 € for the hydrogen storage plant. The amount of investments largely depends on the particular location of the storage plant.

Operation and maintenance (O&M) costs include fixed O&M costs (Table 1) and energy costs for charging the storage plant as shown in the description of mathematical model, Eq. (8).

For the purpose of this study, we assumed that construction of both plants takes two years, electricity market price increases uniformly by 3% each year and discount rate is 7%. The planning horizon considered is 30 years.

Feasibility of PSH and hydrogen storage plan—base scenario

The calculated NPV of storage plants is illustrated in Fig. 9. It can be seen that the hydrogen storage plant with current costs

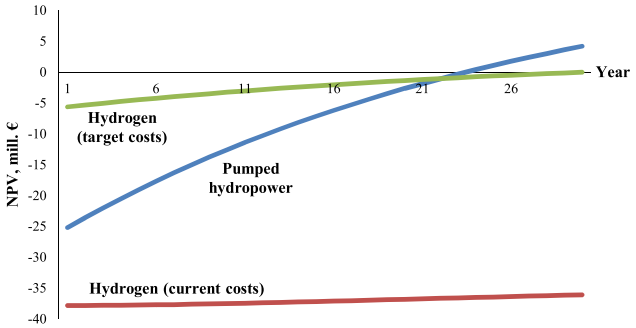


Fig. 9 – NPV of the storage plants of different technologies and varying capital costs.

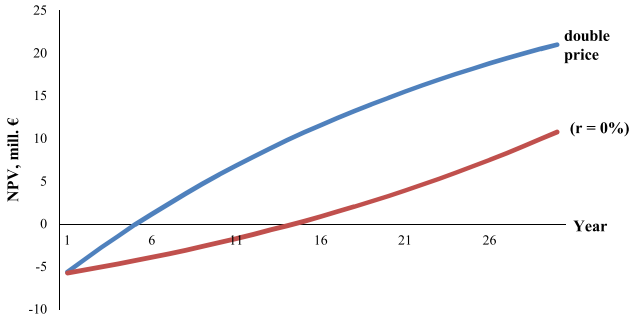


Fig. 11 – NPV of the hydrogen storage plant considering different support schemes.

does not reach a break-even point and its NPV is −32.7 million € at the end of 30-year planning horizon. In contrast, the PSH plant has a payback period of 24 years, its IRR is 8.3% and NPV is 4 204 645 € by year 30.

All the other parameters being the same, the PCS and BOP costs of a closed-loop hydrogen storage plant should be 207.84 €/kW (resulting in TCC of 5 750 915 €) for it to break-even by year 30. In contrast to PSH, hydrogen conversion and storage technologies are not yet mature, so it is reasonable to expect a significant decrease of their capital costs in the following years.

Feasibility of hydrogen storage plant considering additional price volatility and support schemes

Evidently, for a hydrogen storage plant project presented above to be feasible, the capital costs currently are too high and the difference between peak and off-peak prices is too small. However, increased penetration of intermittent RES is expected to contribute to price volatility, especially if not enough large-scale storage projects are implemented. Thereby we investigated several scenarios where electricity market prices do not rise uniformly as assumed in the previous subsection, but rather the difference between the maximum and minimum price increases with each year. For five different scenarios with 0%, 1%, 3%, 5% and 8% increase in price spread, the NPV of the hydrogen facility is illustrated in Fig. 10.

It can be concluded that given current capital cost estimates, at least 8% annual increase in the difference between on-peak and off-peak prices is necessary for the previously described project to achieve payback within its lifetime.

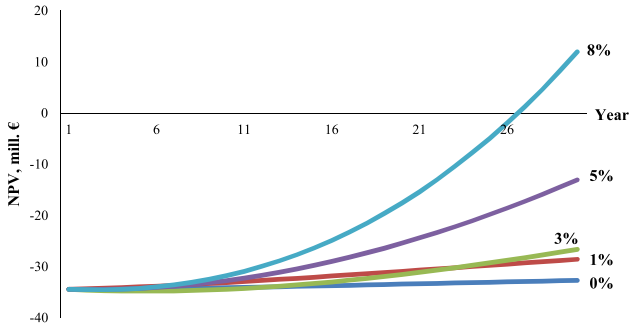


Fig. 10 – NPV of the hydrogen storage plant for various scenarios of price volatility increase.

On the other hand, since the increase of energy storage capacity allows to accommodate a larger amount of intermittent RES in the power system and thus provides overall socio-economic benefits, the government could provide additional incentives for storage plant constructors and/or operators. The growth of installed RES capacity worldwide is largely due to the government support schemes which encourage investments into RES power plants and improve their profitability. Similar supporting incentives could be introduced for hydrogen storage plants.

In Fig. 11 the profitability of hydrogen storage considering two different support schemes is illustrated. For these calculations, the investments in plant construction were calculated considering the target costs established in the previous subsection and the rise of electricity market prices was assumed to be uniform with an annual increase of 3%.

The first support scheme under consideration is doubling the selling price of electrical energy the storage plant produces and injects back into the grid. In this case the payback period of investments is six years and NPV by year 30 is 20 985 894 €.

The other potential support mechanism analyzed was providing cheap loans for construction of new storage plants. Assuming a discount rate $r = 0\%$, the hydrogen storage plant would break-even in 15 years and reach an NPV of 10 762 789 € by year 30.

The results of all the above-mentioned price spread scenarios and support schemes are summarized in Table 2.

Table 2 – Project feasibility indicators of hydrogen storage plant for different scenarios.

	Scenario	NPV, €	IRR	Payback period, years
Annual increase of price spread	0%	−32 692 743	−7.5%	–
	1%	−28 558 280	−2.2%	–
	3%	−26 617 462	0.3%	–
	5%	−13 028 401	4.6%	–
	8%	11 992 266	8.5%	27
	Target costs	0	7.0%	30
Support scheme	Double price	20 985 894	27.6%	6
	Cheap loans	10 762 789	7.0%	15

Discussion

Feasibility assessment of stand-alone storage plant projects using two different technologies and operating under market conditions showed that PSH plant is profitable considering the current costs, whereas hydrogen storage plant needs additional support or suitable market conditions to become profitable. Increase of electricity market price spread or governmental support in form of additional price incentives or decreased interest rate on investments would facilitate the profitability of hydrogen storage plant.

The profitability of hydrogen storage could be also increased by using hydrogen for other purposes, e.g., for combined heat and power generation or as fuel for transportation.

The proposed approach and optimization algorithm can be extended to solve more elaborate problems, e.g., by employing energy storage locally together with wind, solar and other RES power plants or targeting particular needs of a local community with insufficient power supply, transmission congestions or isolated power system. In such a case, energy storage helps to defer or avoid additional investments into the power grid and new generation capacities and is more effective in terms of capital rate of return on investment. Energy storage can also help to reduce electricity price peaks in regions with high volatility of prices.

However, the stochastic optimization problem becomes much more complex in the abovementioned applications because multidimensional stochastic processes (such as prices, wind speed, solar radiation, power demand) need to be considered. We plan to extend the current study in the future to consider the multidimensional stochastic processes as well.

Conclusions

1. The dissemination of RES and distributed generation leads to profitable usage of energy storage units and promotes implementation of new energy storage technologies.
2. The model proposed herein allows to take into account the hourly changes of the power plant operational conditions even for a decades-long planning horizon.
3. The long-term profit optimization problem can be divided into 3 parts: medium-term optimization (to estimate the amount of stored energy); short-term optimization (to estimate the daily profit); and long-term optimization (to calculate the NPV and IRR using Monte Carlo simulations with a relatively small number of trials).
4. The model proposed in our study can be used as a decision support tool for power plant investment planning under electricity market conditions as it allows to consider the hourly-changing stochastic parameters as opposed to the conventional investment planning methods which employ deterministic approach and cannot capture the dynamics of variables.

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